

Power Supply Systems

Part II

Phase 6: Ring Main Distributor

Aim: To learn ring main distributor, interconnector, loop-current method and semester numericals from absolute zero.

Style: Simple English, SN ma'am style, clean diagrams, no overlapping words, every diagram word explained.

Target: Ring main theory, equivalent both-end distributor, KVL method, minimum potential point, voltage calculation and interconnector concept.

Subject: Power Supply Systems

Section: Part II

Phase: 6 – Ring Main Distributor

Level: Absolute zero to semester-question-ready

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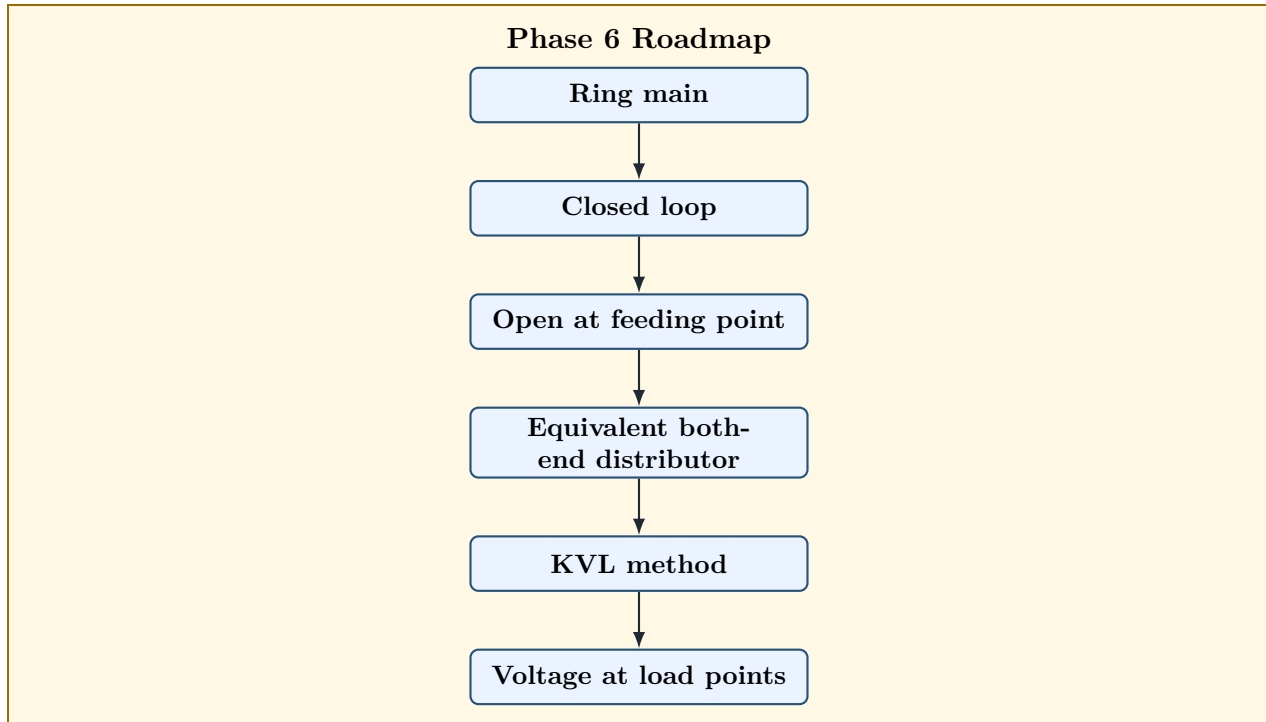
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1 Phase 6 Roadmap

Phase 6 Goal

In Phase 5, we learnt distributor fed at both ends. A ring main distributor is the next step because a ring can be opened at the feeding point and treated like a distributor fed at both ends.



Memory Hook

The main trick:

Ring main problem = Both-end distributor problem after opening the ring at supply point

2 Meaning of Ring Main Distributor

Definition

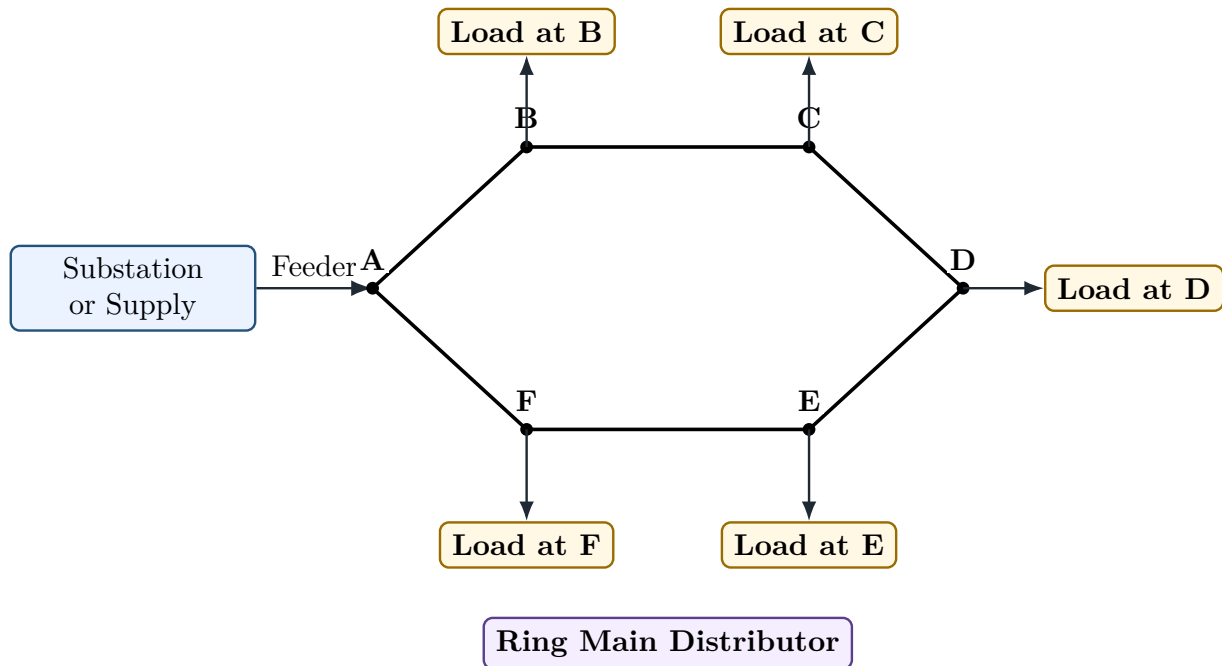
A **ring main distributor** is a distributor arranged in the form of a closed loop and fed at one or more points.

In a ring main system:

- the distributor forms a closed path,
- loads are tapped at different points of the ring,
- current can reach a load from two directions,
- supply reliability is better than a simple radial distributor.

Ring main = closed-loop distributor

3 Basic Ring Main Diagram



How to Read the Diagram

Word/Symbol	Meaning
Substation or Supply	Source from which electrical power enters the ring distributor.
Feeder	Conductor that connects substation to the ring main.
A	Feeding point of the ring. The ring receives supply at this point in this diagram.
B, C, D, E, F	Load tapping points on the ring distributor.
Load at B, C, D, E, F	Consumer loads connected at different points of the ring.
Ring Main Distributor	The complete closed-loop distributor. It has no dead end.
Closed loop shape	Current can reach a load from two possible paths. This improves reliability and voltage regulation.

4 Advantages of Ring Main Distributor

1. It gives better reliability than radial distribution.
2. If one section fails, the faulty section can be isolated.
3. Consumers can still receive supply from the other side.
4. Voltage regulation is better.
5. Current is shared between two paths.
6. It is suitable for urban and important load areas.

Semester Answer Style

Semester line:

Ring main is more reliable because supply can reach loads from two directions.

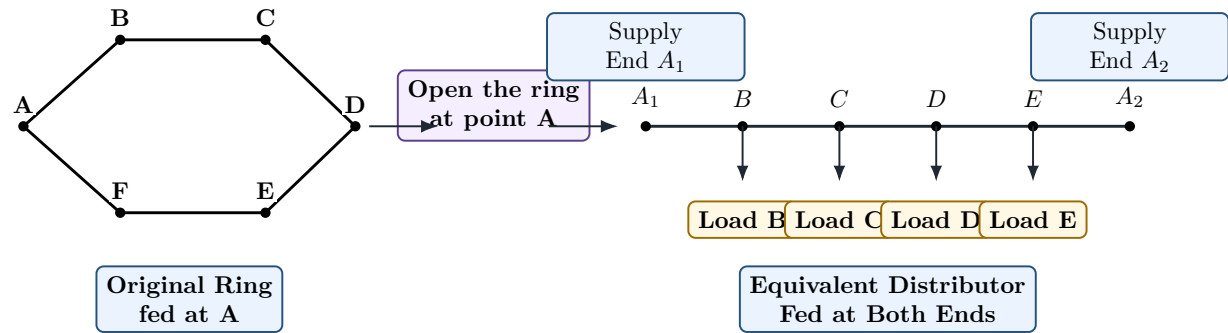
5 Ring Main and Equivalent Both-End Distributor

A ring main fed at point A can be imagined as cut open at A. After opening it at A, the same ring becomes a straight distributor fed at both ends.

Both ends of the opened line are actually the same physical point A. So they have the same voltage.

$V_{A_1} = V_{A_2}$

6 Diagram: Opening a Ring at the Feeding Point



How to Read the Diagram	
Word/Symbol	Meaning
Original Ring fed at A	The real ring distributor before solving. Supply enters at point A.
Open the ring at point A	A mathematical trick. We cut the ring at the supply point to make it a straight line.
A ₁ and A ₂	Two ends of the opened distributor. Physically both represent the same supply point A.
Load B, Load C, Load D, Load E	Loads tapped along the ring after opening it into a line.
Equivalent Distributor Fed at Both Ends	The ring problem can now be solved like a distributor fed at both ends.
Supply End A ₁ , Supply End A ₂	Both ends have the same voltage because both are the original point A.

Memory Hook

For a ring main fed at one point:

Open at supply point $\Rightarrow$ solve as both-end distributor with equal end voltage

7 KVL Loop Method for Ring Main

The most direct method for ring-main numerical is the KVL method.

7.1 Main Idea

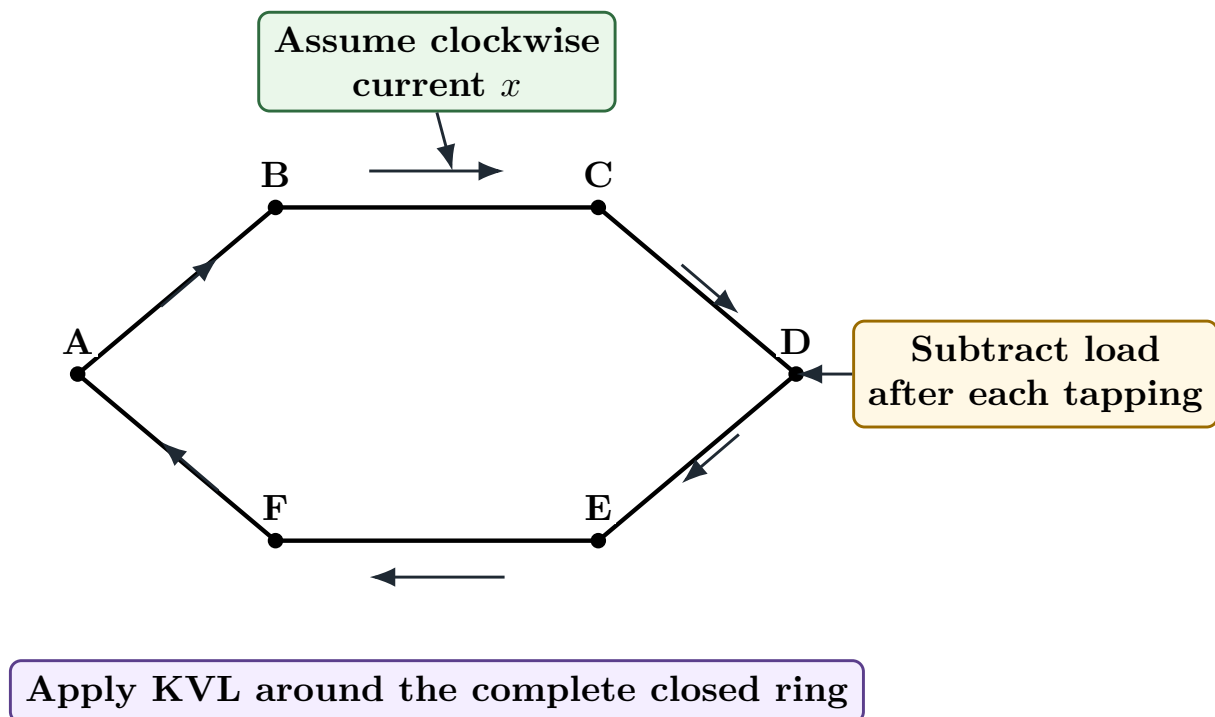
Assume a current x flowing clockwise from the feeding point. After every load point, subtract the load current.

Then apply Kirchhoff's Voltage Law around the complete ring:

$$\sum IR = 0$$

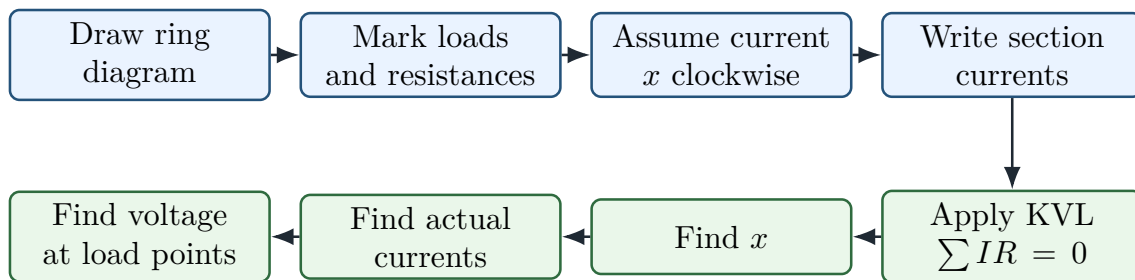
because after completing one full round, we return to the same supply point.

8 KVL Method Diagram



How to Read the Diagram

Word/Symbol	Meaning
Assume clockwise current x	We start the solution by assuming unknown current x flowing around the ring in clockwise direction.
Clockwise arrows	Direction chosen for writing currents and KVL. Actual current may be opposite in some sections.
Subtract load after each tapping	When a load current leaves the distributor, the current in the next section decreases by that load current.
A, B, C, D, E, F	Junction/load points of the ring.
Apply KVL	Around a closed ring, algebraic sum of voltage drops is zero.
Complete closed ring	The path returns to the same point, so starting and ending voltage are the same.

9 Step-by-Step Flowchart for Ring Main Numerical**How to Read the Diagram**

Step	Meaning
Draw ring diagram	Draw the closed-loop distributor and show the supply point.
Mark loads and resistances	Write every load current and every section resistance on the diagram or in a table.
Assume current x clockwise	Choose one unknown current direction to start. Clockwise is usually convenient.
Write section currents	After every load, subtract that load from the previous section current.
Apply KVL $\sum IR = 0$	Add algebraic voltage drops around the complete ring and equate to zero.
Find x	Solve the KVL equation.
Find actual currents	Negative current means actual direction is opposite to assumed direction.
Find voltage at load points	Start from supply voltage and subtract or add section voltage drops carefully.

10 Current Table Pattern for Ring Main

Suppose ring A–B–C–D–A is fed at A. Loads are:

$$I_B, \quad I_C, \quad I_D$$

Assume current x flows clockwise from A to B.

Section	Assumed Current	Meaning
AB	x	Current leaving supply point A in clockwise direction.
BC	$x - I_B$	Load I_B is tapped at B, so current decreases.
CD	$x - I_B - I_C$	Loads at B and C have already been tapped.
DA	$x - I_B - I_C - I_D$	All loads have been tapped before returning to A.

$$xR_{AB} + (x - I_B)R_{BC} + (x - I_B - I_C)R_{CD} + (x - I_B - I_C - I_D)R_{DA} = 0$$

This is the master KVL equation for a simple ring main.

11 Minimum Potential Point in Ring Main

Definition

The **minimum potential point** in a ring main is the point where the current changes direction.

In ring-main numerical:

- after solving x , find section currents,
- if a section current becomes negative, actual current is opposite,
- the point where current reverses is the minimum potential point.

$$\text{Minimum potential point} = \text{point where current changes direction}$$

12 Solved Example 1: Ring Main by KVL Method

12.1 Problem

A 2-wire DC ring distributor A–B–C–D–A is fed at point A at 250 V. Loads are:

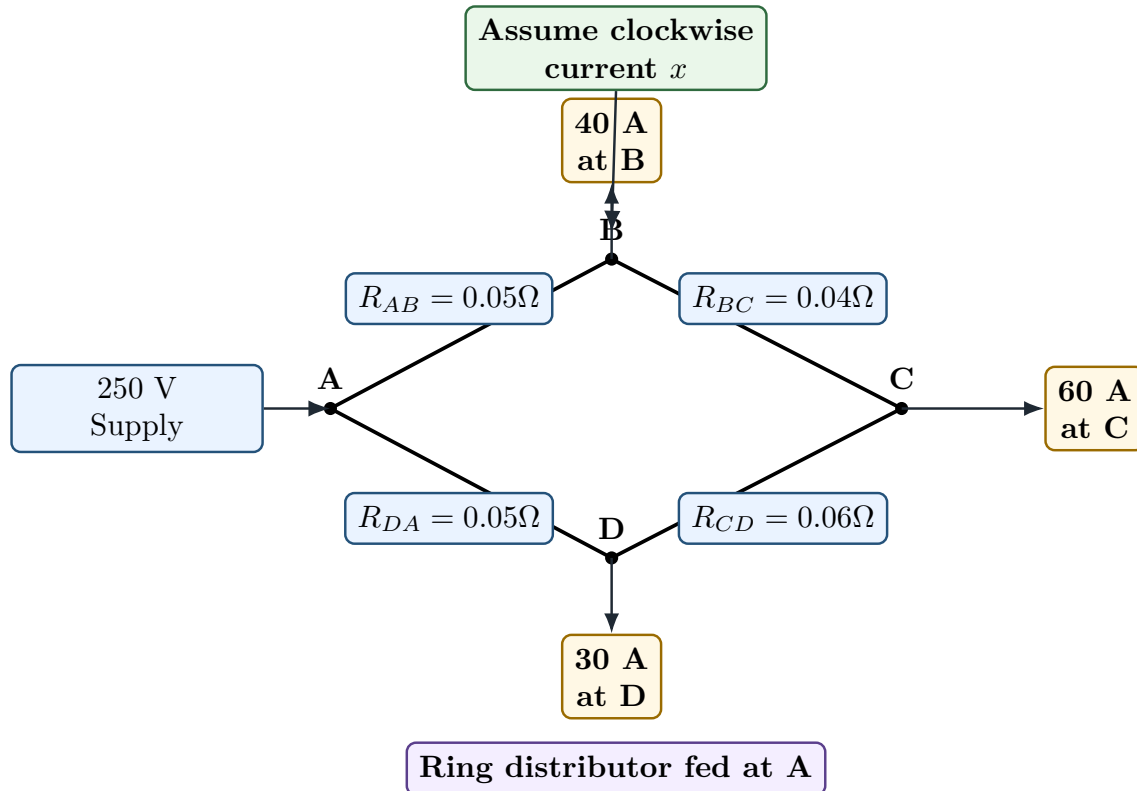
$$40 \text{ A at B, } 60 \text{ A at C, } 30 \text{ A at D}$$

Section resistances for the complete 2-wire distributor are:

$$R_{AB} = 0.05\Omega, \quad R_{BC} = 0.04\Omega, \quad R_{CD} = 0.06\Omega, \quad R_{DA} = 0.05\Omega$$

Find current in each section and voltage at each load point.

12.2 Clean Ring Diagram



How to Read the Diagram

Word/Symbol	Meaning
250 V Supply	Voltage at the feeding point A.
A	Feeding point of the ring distributor.
B, C, D	Load tapping points on the ring.
40 A at B	Load current taken from point B.
60 A at C	Load current taken from point C.
30 A at D	Load current taken from point D.
$R_{AB}, R_{BC}, R_{CD}, R_{DA}$	Resistances of the four ring sections. They are given for the complete 2-wire path.
Assume clockwise current x	We first assume unknown current x flowing A to B to C to D to A.
Ring distributor fed at A	Closed loop distributor receiving supply at A.

12.3 Step 1: Assume Current

Assume current x flows clockwise from A to B.

Then section currents are:

$$I_{AB} = x$$

$$I_{BC} = x - 40$$

$$I_{CD} = x - 40 - 60 = x - 100$$

$$I_{DA} = x - 40 - 60 - 30 = x - 130$$

12.4 Step 2: Apply KVL Around the Ring

$$x(0.05) + (x - 40)(0.04) + (x - 100)(0.06) + (x - 130)(0.05) = 0$$

Expand:

$$0.05x + 0.04x - 1.6 + 0.06x - 6 + 0.05x - 6.5 = 0$$

$$0.20x - 14.1 = 0$$

$$0.20x = 14.1$$

$$\boxed{x = 70.5 \text{ A}}$$

12.5 Step 3: Section Currents

$$I_{AB} = x = 70.5 \text{ A}$$

$$I_{BC} = x - 40 = 70.5 - 40 = 30.5 \text{ A}$$

$$I_{CD} = x - 100 = 70.5 - 100 = -29.5 \text{ A}$$

$$I_{DA} = x - 130 = 70.5 - 130 = -59.5 \text{ A}$$

Negative sign means actual current is opposite to the assumed clockwise direction.

Therefore:

$$\boxed{I_{AB} = 70.5 \text{ A from A to B}}$$

$$\boxed{I_{BC} = 30.5 \text{ A from B to C}}$$

$$\boxed{I_{CD} = 29.5 \text{ A from D to C}}$$

$$\boxed{I_{AD} = 59.5 \text{ A from A to D}}$$

12.6 Step 4: Locate Minimum Potential Point

Current in BC is from B to C. Current in CD is actually from D to C.
So current comes to point C from both sides.

Minimum potential point is C.

12.7 Step 5: Voltage at Load Points

Voltage at A:

$$V_A = 250 \text{ V}$$

Voltage drop from A to B:

$$\Delta V_{AB} = 70.5 \times 0.05 = 3.525 \text{ V}$$

$$V_B = 250 - 3.525 = 246.475 \text{ V}$$

Voltage drop from B to C:

$$\Delta V_{BC} = 30.5 \times 0.04 = 1.22 \text{ V}$$

$$V_C = 246.475 - 1.22 = 245.255 \text{ V}$$

Voltage drop from A to D along AD:

$$\Delta V_{AD} = 59.5 \times 0.05 = 2.975 \text{ V}$$

$$V_D = 250 - 2.975 = 247.025 \text{ V}$$

$$V_B = 246.475 \text{ V}$$

$$V_C = 245.255 \text{ V}$$

$$V_D = 247.025 \text{ V}$$

$$V_{\min} = 245.255 \text{ V at C}$$

Semester Answer Style

The voltage at C is the minimum because current from both sides meets at C.

13 Equivalent Both-End Method for the Same Example

The same problem can also be solved by opening the ring at A.

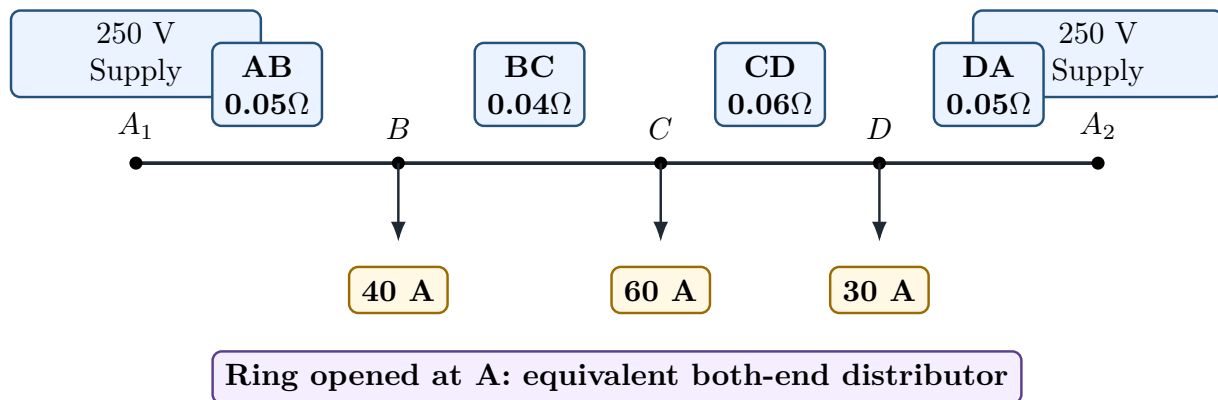
$$A \rightarrow B \rightarrow C \rightarrow D \rightarrow A$$

becomes:

$$A_1 \rightarrow B \rightarrow C \rightarrow D \rightarrow A_2$$

where:

$$V_{A_1} = V_{A_2} = 250 \text{ V}$$



How to Read the Diagram

Word/Symbol	Meaning
A_1 and A_2	Two ends obtained by opening the original ring at A. Both are physically the same supply point.
250 V Supply at both ends	Since A_1 and A_2 are the original point A, both ends have the same voltage.
B, C, D	Load points in the opened straight distributor.
AB, BC, CD, DA	Sections of the original ring, now written in straight-line order.
Equivalent both-end distributor	This is not a different physical system. It is only a solving method.

14 Ring Main with Interconnector

Definition

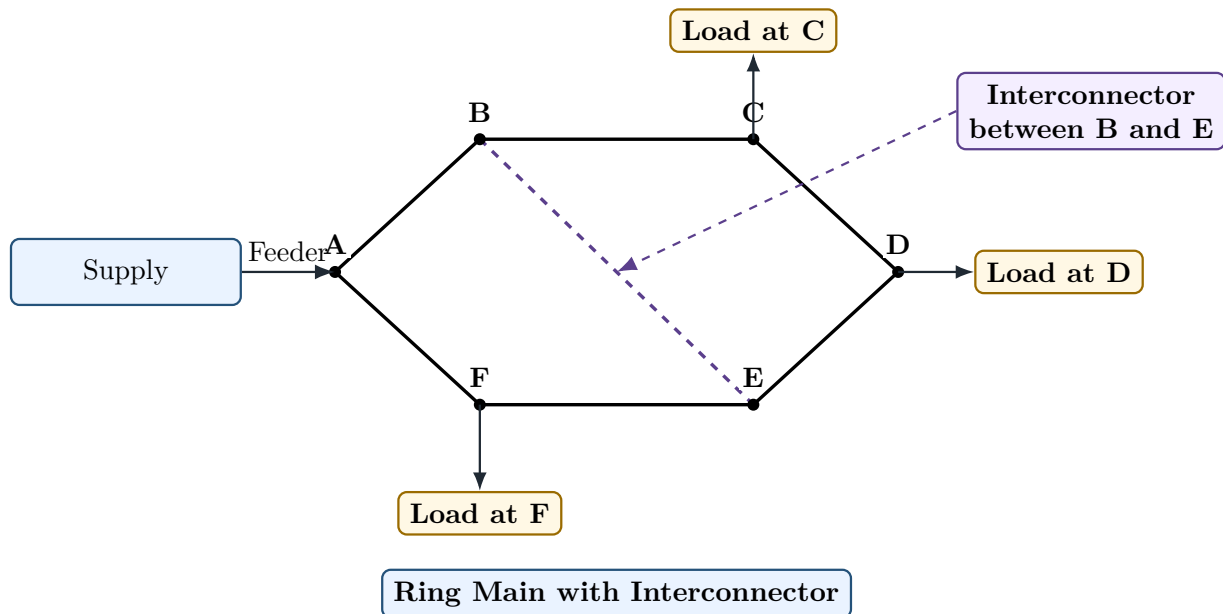
A **ring main with interconnector** is a ring main system in which two non-adjacent points of the ring are connected by an extra conductor called an interconnector.

The interconnector:

- provides an additional path for current,
- improves voltage regulation,

- improves reliability,
- helps load sharing,
- reduces voltage drop in many cases.

15 Diagram: Ring Main with Interconnector



How to Read the Diagram

Word/Symbol	Meaning
Ring Main	Closed-loop distributor around the load area.
Interconnector between B and E	Extra conductor connecting point B and point E. It gives an additional current path.
Dashed line	The interconnector is shown dashed only to distinguish it from the outer ring. It is still an electrical conductor.
Supply	Feeding point of the distribution network.
Feeder	Conductor bringing power from the substation to the ring.
Load at C, D, F	Consumers connected at different points of the ring.
A, B, C, D, E, F	Junction points of the ring main.

16 How to Solve Ring Main with Interconnector

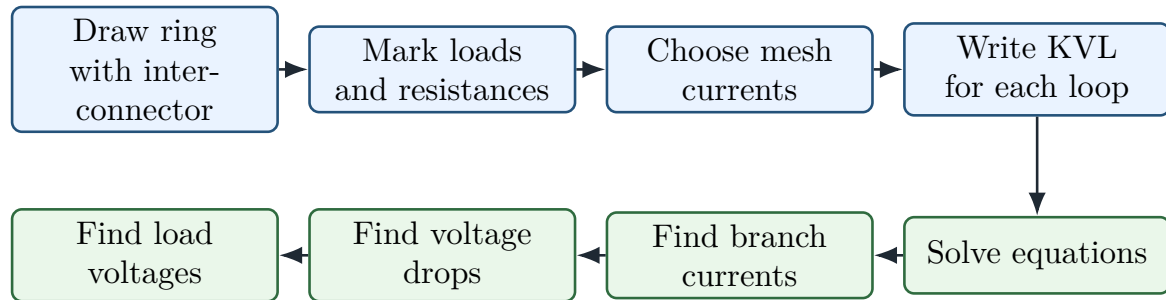
Ring main with interconnector is a network problem. It may require two independent loop equations.

16.1 Main Steps

1. Draw the ring and interconnector clearly.
2. Mark all loads and section resistances.
3. Assume mesh currents in independent loops.

4. Write KVL equation for each loop.
5. Solve simultaneous equations.
6. Find actual branch currents.
7. Calculate voltage drops and load-point voltages.

17 Flowchart for Ring Main with Interconnector



How to Read the Diagram

Step	Meaning
Draw ring with inter-connector	Draw the outer closed ring and the inner connecting conductor separately.
Mark loads and resistances	Every branch must have known load or resistance value before equation writing.
Choose mesh currents	Assume loop currents in the independent loops of the network.
Write KVL for each loop	Each independent loop gives one equation.
Solve equations	Solve the simultaneous equations to obtain mesh currents.
Find branch currents	Convert mesh currents into actual branch currents.
Find voltage drops	Use IR drop in each section.
Find load voltages	Subtract voltage drops from supply voltage along suitable paths.

18 Comparison: Ring Main and Radial Distributor

Point	Radial Distributor	Ring Main Distributor
Shape	Open-ended distributor	Closed-loop distributor
Supply path	One path only	Two possible paths
Reliability	Lower	Higher
Voltage regulation	Poorer at far end	Better due to current sharing
Fault effect	Consumers beyond fault may lose supply	Faulty section can be isolated
Cost	Lower	Higher

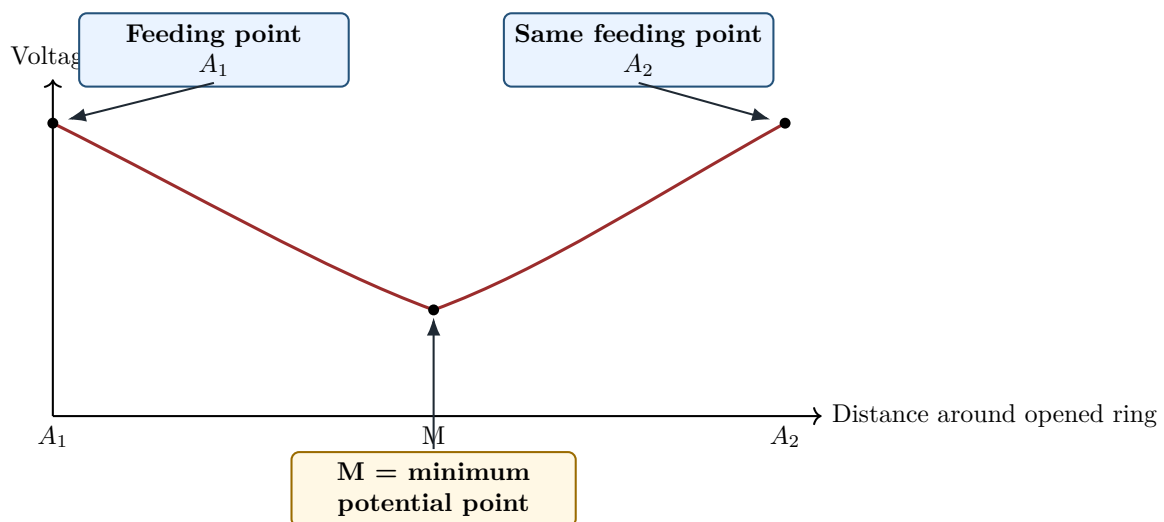
Use	Small or rural areas	Urban and important load areas
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19 Comparison: Ring Main and Both-End Distributor

Point	Both-End Distributor	Ring Main Distributor
Physical shape	Straight distributor fed at two ends	Closed-loop distributor
Mathematical relation	Directly solved as both-end fed line	Can be opened at feed point and solved as both-end fed line
End voltages	May be equal or unequal	If opened at same feeding point, end voltages are equal
Current direction	From both ends towards minimum point	Current flows through two paths around the ring
Minimum voltage	At current reversal point	At current reversal point/load point
Use in calculation	Basic method	Application of both-end method or KVL loop method

20 Voltage Profile of a Ring Main

In a ring main fed at one point, voltage is maximum at the feeding point. Voltage falls along both directions from the feeding point. The minimum voltage occurs at the point where currents from both directions meet.



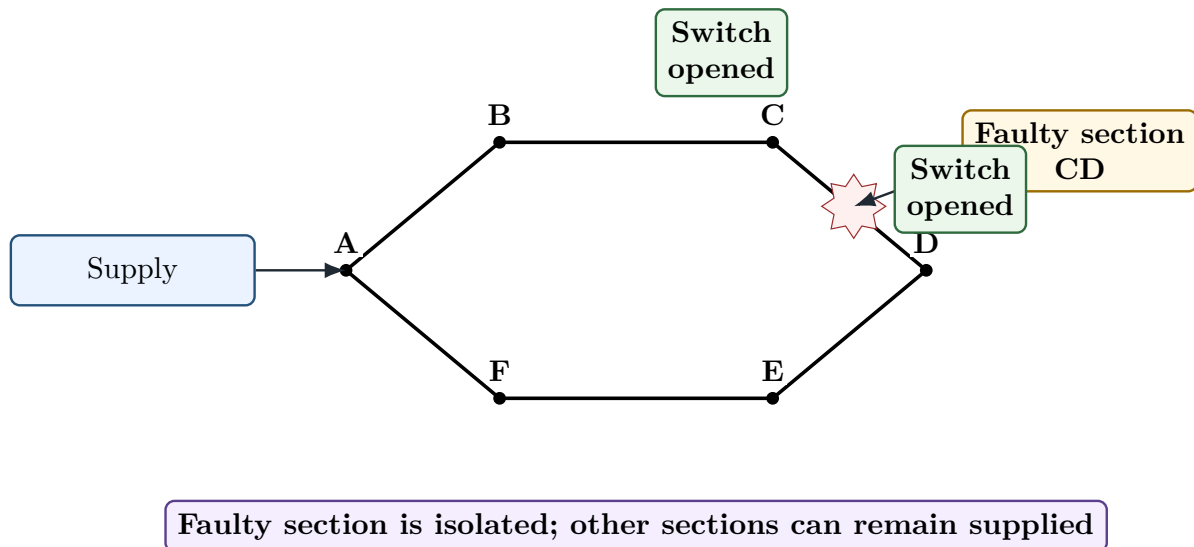
How to Read the Diagram

Word/Symbol	Meaning
Distance around opened ring	The ring has been opened at the feeding point and drawn as a straight line.
Voltage	Voltage available at each point of the opened ring.
A_1 and A_2	Two ends obtained by opening the same feeding point A. Their voltages are equal.
M	Minimum potential point. It is where current from both directions meets.
Voltage curve	Shows voltage falling from both sides towards M.
Feeding point A_1 , A_2	Same physical point A, shown twice after opening the ring.

21 Protection and Switching Idea in Ring Main

Ring main is useful because the faulty section can be isolated by opening switches on both sides of the fault.

21.1 Fault Isolation Diagram



How to Read the Diagram

Word/Symbol	Meaning
Faulty section CD	The conductor section between C and D has a fault.
Switch opened	Switches on both sides of the faulty section are opened to isolate it.
Supply	Source feeding the ring main.
Other sections	Sections not under fault can still receive supply through the remaining part of the ring.
Fault isolation	The process of disconnecting only the faulty part instead of shutting down the whole distributor.

22 Reverse Engineering of Semester Questions

Syllabus Word	Concept	Semester Question Form
Ring main	Closed-loop distributor	Explain ring-main distributor with diagram.
Comparison of distribution systems	Radial vs ring main	Why is ring main better than radial system?
Current and voltage profile	Current reversal and minimum voltage	Find minimum potential point and voltage.
Ring distributor numerical	KVL around closed loop	Solve ring main distributor and find voltage at load points.
Interconnector	Extra connection between two points	Explain ring main with interconnector and its advantages.
Fault isolation	Reliability improvement	Explain how ring main maintains supply during fault.
Equivalent distributor	Open ring at feeding point	Show that ring distributor can be solved as both-end fed distributor.

23 Most Predictable Semester Questions from Phase 6

No.	Question	Marks
1	What is ring main distributor? Explain with neat diagram.	4–5
2	What are the advantages of ring main distributor over radial distributor?	4–6
3	Explain how a ring main distributor can be converted into an equivalent distributor fed at both ends.	5
4	A ring main distributor is fed at one point and has loads at different points. Find section currents and voltages.	8–10
5	Explain the KVL loop method for solving ring distributor problems.	5

6	What is minimum potential point in a ring main? How is it located?	4
7	Explain ring main with interconnector and mention its advantages.	5–6
8	Draw voltage profile of a ring main distributor and explain.	4
9	Explain how faulty section can be isolated in ring main system.	4
10	Compare radial, ring main and interconnected distribution systems.	6

24 Exam Answer Templates

24.1 Template 1: Theory Answer on Ring Main

Semester Answer Style

Answer sequence:

1. Define ring main distributor.
2. Draw clean closed-loop diagram.
3. Explain supply point, feeder, load points and closed loop.
4. Write advantages.
5. Mention current can reach loads from two directions.
6. Conclude that reliability and voltage regulation are better than radial system.

24.2 Template 2: Ring Main Numerical by KVL

Semester Answer Style

Answer sequence:

1. Draw the ring with loads and resistances.
2. Assume clockwise current x .
3. Write section currents after subtracting loads.
4. Apply:

$$\sum IR = 0$$

5. Solve for x .
6. Convert negative currents into actual opposite directions.
7. Locate current reversal point.
8. Calculate voltages at load points.

24.3 Template 3: Ring Main with Interconnector

Semester Answer Style

Answer sequence:

1. Define interconnector.
2. Draw ring main with an additional dashed interconnector.
3. Explain that it provides extra current path.
4. Write advantages: reliability, voltage regulation, load sharing.
5. For numerical, use mesh-current method and KVL equations.

25 Phase 6 Final Memory Sheet

One-Page Memory Sheet

1. Ring main distributor is a closed-loop distributor.
2. Loads can receive supply from two directions.
3. Ring main has better reliability than radial distributor.
4. A ring fed at one point can be opened at that point and solved as a distributor fed at both ends.
5. In KVL method, assume clockwise current x .
6. After every load, subtract the load current from section current.
7. Around the ring:

$$\sum IR = 0$$
8. Negative section current means actual direction is opposite.
9. Minimum potential point is where current changes direction.
10. Interconnector is an extra conductor connecting two non-adjacent points of the ring.
11. Interconnector improves voltage regulation, reliability and load sharing.
12. Faulty section of ring can be isolated by opening switches on both sides of the fault.

26 Phase 6 Self-Test

Answer these without seeing notes:

1. What is a ring main distributor?
2. Why is ring main better than radial distributor?
3. How can a ring main be converted into an equivalent both-end distributor?
4. Why are the two ends of the opened ring at equal voltage?
5. What is the KVL method for ring distributor?
6. If clockwise current is x , how do section currents change after each load?
7. What does negative section current mean?
8. How do you locate minimum potential point?

9. What is an interconnector?
10. Why does interconnector improve reliability?
11. How can a faulty section be isolated in a ring main?

Common Mistake

Do not put labels inside the ring while drawing. Keep labels outside and connect them using arrows. This keeps the diagram clean and semester-friendly.

End of Phase 6

Next Phase:

Phase 7 – Three-Wire DC System and Unbalanced Loads

DC 3-wire system, neutral current, balancer set, 3-phase 4-wire unbalanced load and neutral current calculation.